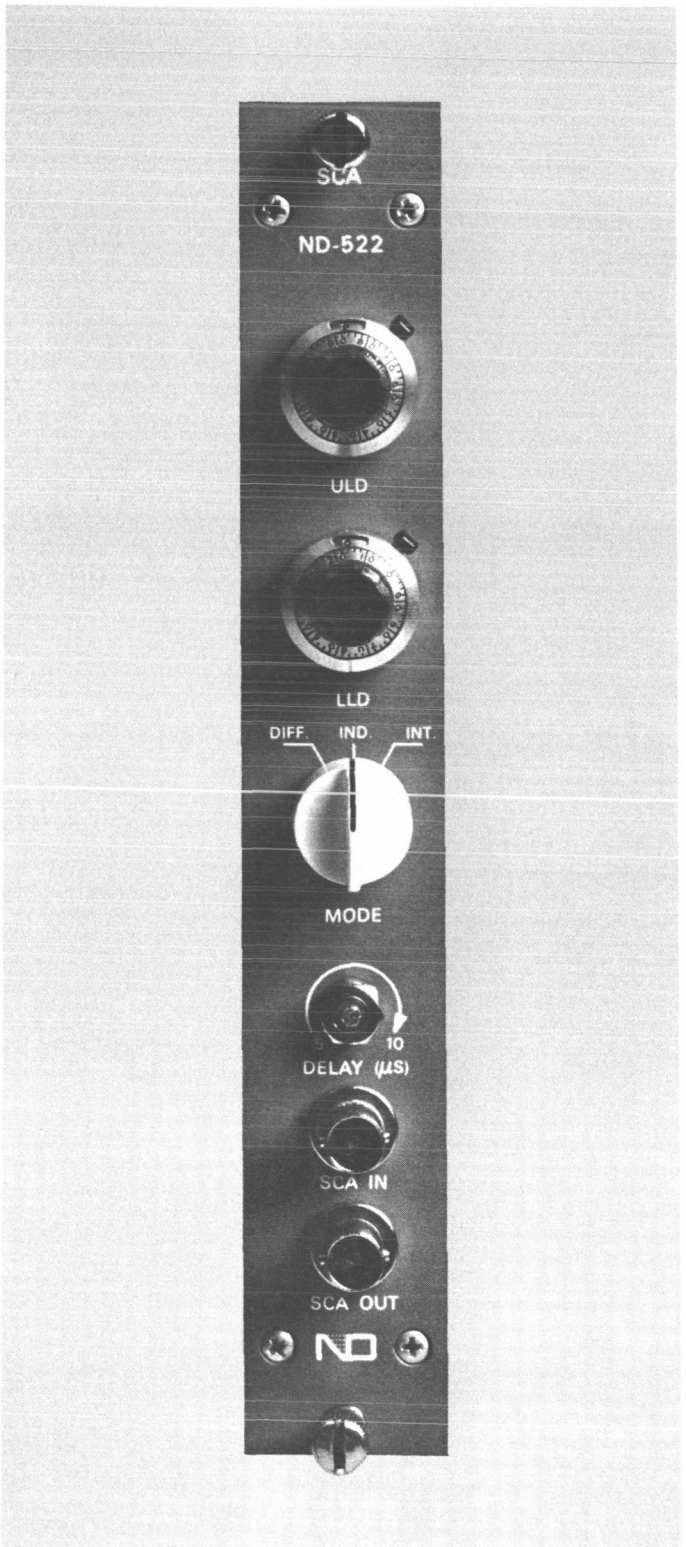
TECHNICAL SPECIFICATIONS

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ND-522 SCA SINGLE CHANNEL ANALYZER

The ND-522 Single Channel Analyzer module generates an output whenever a linear input signal satisfies the pulse height requirements imposed by the LOWER LEVEL and UPPER LEVEL discriminators. These discriminators may be operated in one of three modes, as determined by the front panel Mode switch, and their discrimination level is adjusted by two ten-turn front panel controls. Provisions for timing the occurence of an output signal by either internally generated strobe pulses, externally applied strobe pulses, or an internally generated reset with externally applied strobe pulses for coincidence timing, are provided by means of a Strobe mode switch. The strobing capabilities provided by this analyzer enable it to be used in a wide variety of coincidence or anti-coincidence gating systems.

A unique feature of the ND-522 SCA is the variable output delay timing obtained with internal strobe operation. The front panel Delay control allows the operator to delay the internal strobe pulse from 0.5 to 10 microseconds with respect to the onset of leading edge of discriminator triggering. In coincidence gating systems, where linear signals (up to 10 microseconds rise and fall time constants) must be handled, no additional linear signal delay circuits are required since input pulse width imposes no restrictions on discriminator triggering.

Single Channel Analyzer Input

Signal Requirements Front panel BNC accepts positive monopolar or initially positive bi-polar amplifier outputs to ±15 volts. Input rise times .05 to 10 microseconds with less than 0.1% sensitivity change.

Input Impedance 1,000 ohms, A.C. coupled, long time constant. D.C. coupling is recommended for bipolar pulse and pulses greater than +12 volts.

Mode Switch

Integral Upper Level Discriminator (delta E) is disabled. The only criteria for pulse acceptance being an amplitude exceeding the Lower Level Discriminator threshold (E) setting.

Independent Both E and delta E discriminators operate separately. Pulse acceptance occurs when the pulse amplitude lies between E and delta E settings.

Differential Delta E operates with respect to E. Pulse acceptance occurs when the pulse amplitude lies between E and delta E.

Lower Level Discriminator

Range From 100 millivolts to 10 volts in all three operating modes.

Integral Linearity 0.3% over the range from 100 millivolts to 10 volts.

Upper Level Discriminator

Range From 100 millivolts to 10 volts in the Independent Mode and from E to E +2 volts (i.e., 0% to 20% window) in the Differential Mode.

Integral Linearity 0.3% for both Independent and Differential Modes.

Differential Linearity Does not exceed 1% of window width.

Internal Strobe Discriminator

Generates a strobe pulse when input signal exceeds its threshold. Internal Strobe Discriminator Threshold is referenced to the Lower Level Discriminator Threshold; an internal trimpot provides continuous control of this bias from 0.35 E to E. A front panel delay control enables the actual strobe operation to occur at any time from 0.5 to 10 usec after Internal Strobe Discriminator triggering.

Strobe Switch

Internal Internal strobe discriminator determines timing of SCA output signal in relationship to the setting on the front panel Delay control.

External When in External strobe an SCA output signal is generated after a suitable strobe pulse is applied. The strobe pulse may be applied at any time after the E discriminator triggers.

Internal Reset Internal strobe discriminator resets the E and delta E discriminators as in normal Internal strobe operation, but the single channel analyzer anticoincidence gate must be strobed with an external pulse within the time period set on the front panel Delay control, if an SCA output signal is to be generated. Delays up to 30 microseconds may be obtained with additional capacitance.

External Strobe Input

Signal Requirements Rear panel BNC accepts a 3 volt to 10 volt positive pulse.

Impedance Approximately 400 ohms, A.C. coupled.

Pulse Shape Minimum width 20 nanoseconds. Leading edge used for timing and should be fast, i.e. 10-20 nanoseconds rise time, for minimum single channel output time jitter.

Single Channel Analyzer Output

Signal Characteristics Front panel BNC delivers a 10 volt positive rectangular pulse.

Impedance 50 ohms, A. C. coupled.

Pulse Shape 300 nanoseconds FWHM; 15 nanoseconds rise time, 15 nanoseconds fall time.

Pulse Pair Resolution

Approximately 0.6 microseconds in internal strobe.

Coincidence Output

Signal Characteristics Rear panel BNC provides a 6 volt positive rectangular pulse when D.C. coupled to the COINC/ANTI-COINC input of either a Series 2200 or 3300 Nuclear Data analyzer. A 2 volt positive rectangular pulse may be obtained by changing the position of two wires.

Pulse Shape Nominally 6 microseconds FWHM, a single component change allows output pulse widths from 1 to 10 microseconds.

Anti-Coincidence Output

Signal Characteristics Rear panel BNC provides a 6 volt negative rectangular pulse when D.C. coupled to the COINC/ANTI-COINC input of either a Series 2200 or 3300 Nuclear Data analyzer. A 2 volt negative rectangular pulse may be obtained by changing the position of two wires.

Pulse Shape Nominally 6 microseconds FWHM, a single component change allows output pulse widths from 1 to 10 microseconds.

General

Power Requirements +24 vdc at 25 ma
+12 vdc at 200 ma
−12 vdc at 50 ma
−24 vdc at 3 ma

Size 8.71" high x 1.34" wide x 9.7" deep (standard AEC single width).

Weight Approximately 2 pounds.

Part Number

74-0110